PROJECT DOCUMENT

BIOMASS ENERGY & AI

Reforesting in Mitsue

Founding Story

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A Forest, a Community Center, and a Future Powered by Itself

> **"The forest our ancestors planted — the power that sustains the village they built"** 先人が植えた森が、今、村を支える力になる。

A Village in the Mountains of Nara

Mitsue sits deep in the mountains of Nara Prefecture — a quiet village of cedar slopes, clear water, and an aging community. Like hundreds of villages across rural Japan, it has watched its young people leave for the cities, its school close, and its forests grow heavy with sugi cedar planted in a different era.

The **former Sugano Elementary School (Mitsue Taiken Koryukan)** — a community exchange center, the leading candidate site (final choice confirmed in Phase 1) — stands available: a versatile space with room for a bold new purpose.

We believe it can become something remarkable.

A Simple Idea, Carefully Built

The BIOMASS ENERGY & AI project is a 25-year initiative to transform the former Sugano Elementary School (Koryukan) or a disused village factory building — the former Sugano school is the leading candidate; final site confirmed in Phase 1 — into a living example of how rural Japan can power its own future.

We will:

- **Restore native forests** in place of aging sugi monoculture, healing land that has been ecologically depleted for generations. Thinning the overgrown sugi both heals the forest and generates the fuel that powers the village.
- **Generate electricity and heat from sugi forest thinnings** through a biomass combined heat and power (CHP) system — a circular energy economy where the forest powers the village. Electricity is the primary output and heat the secondary; complementary solar panels and EV charging stations round out a locally owned system that keeps the village powered during blackouts and ready for the coming EV transition. Battery storage to be evaluated in the feasibility study.
- **Build a small, sustainable AI data center** in the former Sugano school (Koryukan) or a disused village factory — the leading candidate is the former Sugano school; disused factories are the alternative; final site confirmed in Phase 1 — powered entirely by local energy, providing digital infrastructure where it has never existed before.

Three things, woven together: ecology, energy, and the digital age.

Why 25 Years

We are not building for the short term. The forests we plant will mature over decades. The technology we install will evolve. By the time our project completes its first full cycle, **localized small-scale fusion energy** may finally be commercially available — and the biomass CHP, solar, and EV infrastructure we build now will have served as the bridge.

In the meantime, Japan's vehicle fleet is gradually shifting toward electrification. New plug-in vehicle sales are projected to reach 30–40% by the mid-2030s — but the total share of EVs on the road is expected to remain around 10–15% through 2036. Mitsue is not waiting for EV ownership to become widespread. By building EV-capable charging infrastructure today, the village positions itself as forward-looking and EV-ready: serving the EV owners already here, welcoming visitors, and prepared for the longer transition ahead.

The BIOMASS ENERGY & AI project is designed for that future.

Built on Citizen Science and Open Values

The project grows out of more than two decades of work with **Safecast**, the global open citizen science network for environmental monitoring, founded after the 2011 Fukushima disaster. The same values that made Safecast possible — open data, community ownership, and patient long-term commitment — shape everything about the BIOMASS ENERGY & AI project.

All environmental, energy, and forestry data from the project will be **published openly**. Any village, anywhere in the world, will be able to learn from what we do here.

The People Involved

The project is led by **Rob Oudendijk**, a Dutch electrical engineer born in the Netherlands — the same country that is home to ASML, where innovation is natural — who has lived in Mitsue since 2012 and is a core hardware developer for Safecast.

Advising and supporting the project are:

- Ray Ozzie — Executive Chair, Blues
- **San Poisson** — Project Manager
- **Takuo Dome** — Professor Emeritus, Osaka University · Director, Inochi Forum
- **Evin Zoet** — Co-Representative Director, Transom
- **Yoshiko Zoet-Suzuki** — Co-Representative Director, Transom

A dedicated **non-profit organization** is being established to coordinate the project, bringing together local residents, village leaders, forestry experts, technology partners, and academic collaborators.

A Model, Not a One-Off

Mitsue is not the only village facing these challenges. Across Japan — and across rural communities worldwide — closed schools, aging forests, and shrinking populations are part of the same story.

By documenting everything we do as an open, replicable playbook, we hope the BIOMASS ENERGY & AI project becomes a starting point rather than an endpoint. If it works here, it can work elsewhere.

The Long Walk Begins

This is a project that begins with conversations — with neighbors, landowners, village leaders, and partners. It will be built slowly, carefully, and with deep respect for the community that has called these mountains home for generations.

We invite anyone who shares this vision to walk with us.

The BIOMASS ENERGY & AI project Mitsue Village, Nara Prefecture, Japan